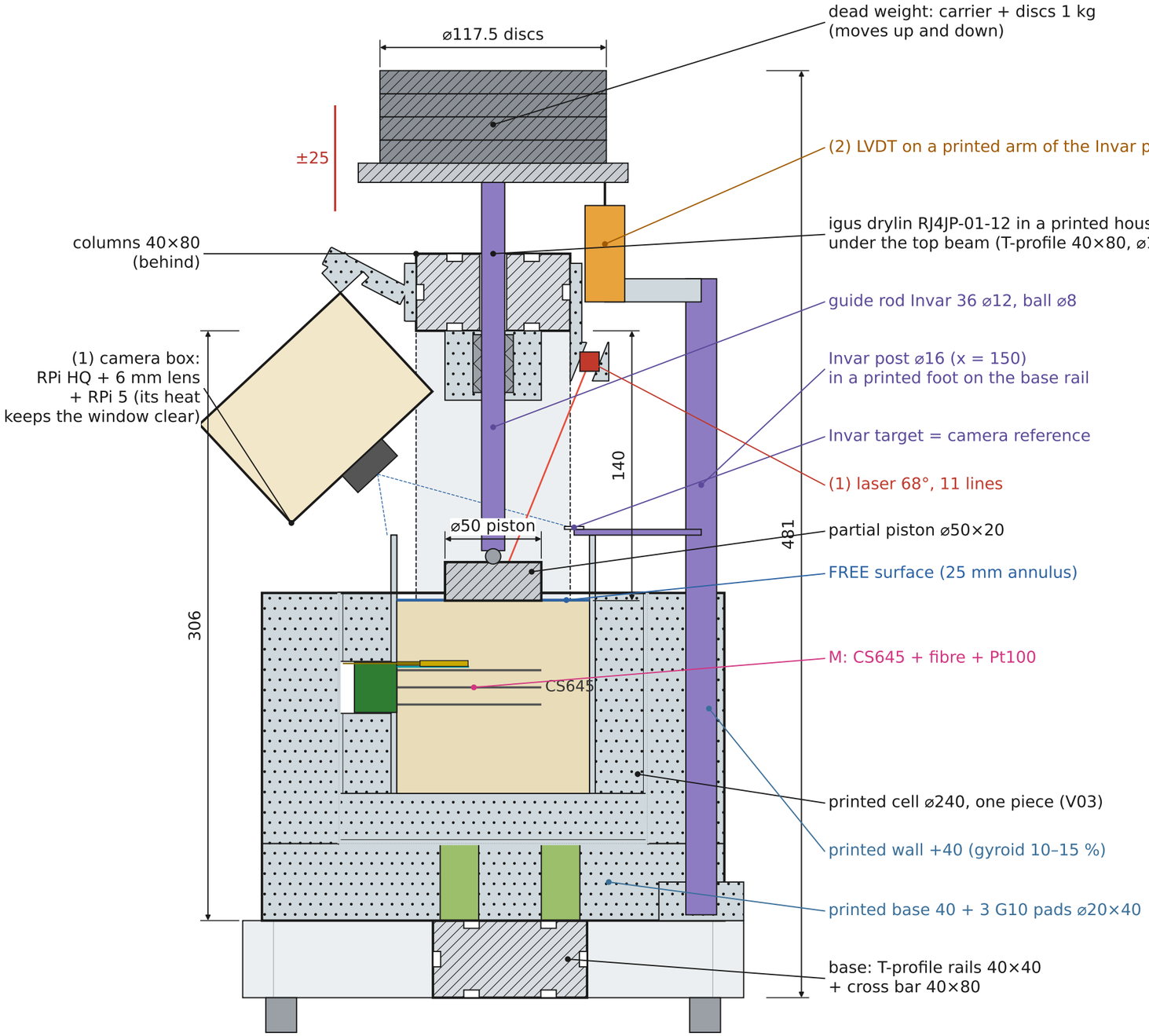


Load under the piston (ø50 = 1963 mm²)

kPa	total kg	1 kg discs	trim kg
5	1.00	0	0.31
10	2.00	1	0.31
15	3.00	2	0.31
20	4.00	3	0.31
25	5.00	4	0.31

moving part without discs ≈ 0.69 kg ≈ 3.5 kPa (minimum)

Parts

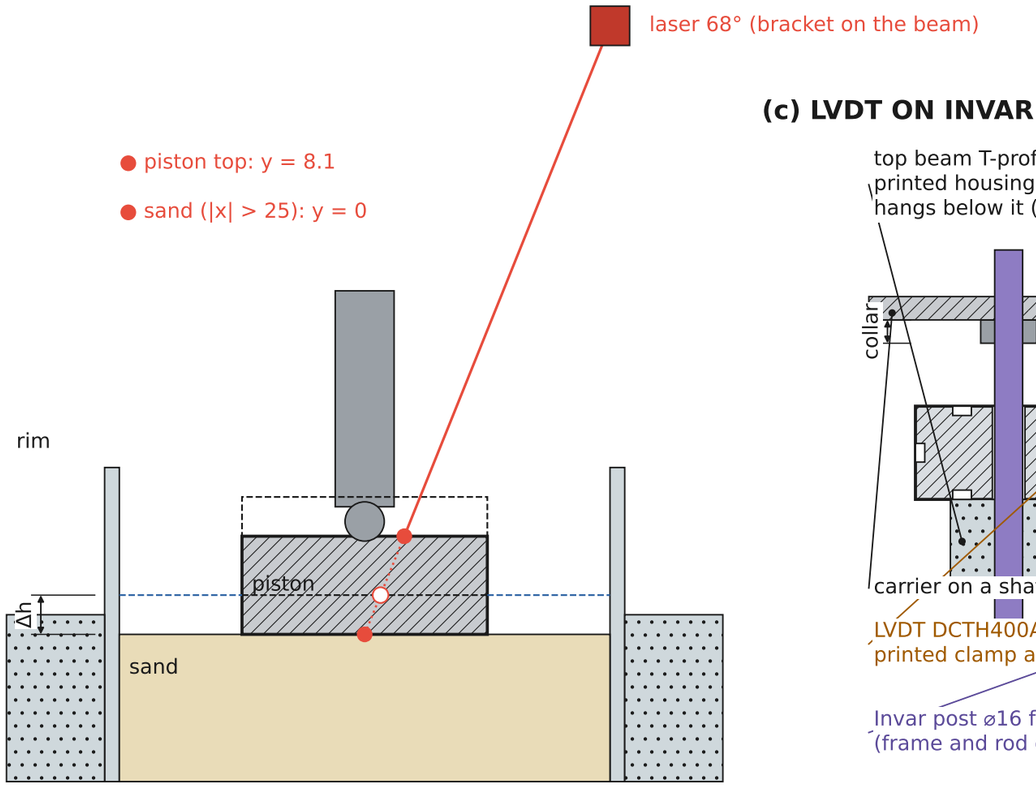
#	part
1	base: 2 rails 40×40 × 260 + cross bar 40×80 × 260, 4 levelling feet M8
2	2 columns 40×80 × 306; top beam 40×80 × 340 (ø14); 12 angle brackets
3	igus RJ4JP-01-12 in a printed housing; Invar rod ø12
4	piston ø50×20 = 2 laser-cut Al discs, ball 5/16"
5	carrier Al ø140×10 on a shaft collar; discs ø117.5×12 = 1 kg
6	printed cell ø240 × 200, ONE piece incl. insulation (V03)
7	sand ø100×100; 3 G10 pads ø20×40 in the base pockets
8	Invar post ø16: LVDT arm + camera target arm
9	(1) camera + RPi 5 box, 11-line laser (V02)
10	(2) LVDT RDP DCTH400AG (V02)
11	point M: CS645, Pt100, optical fibre (existing)

side view y–z through the axis; the T-profile frame only GUIDES the load

Rig C rev. B3 – dead weight on a partial piston, free surface

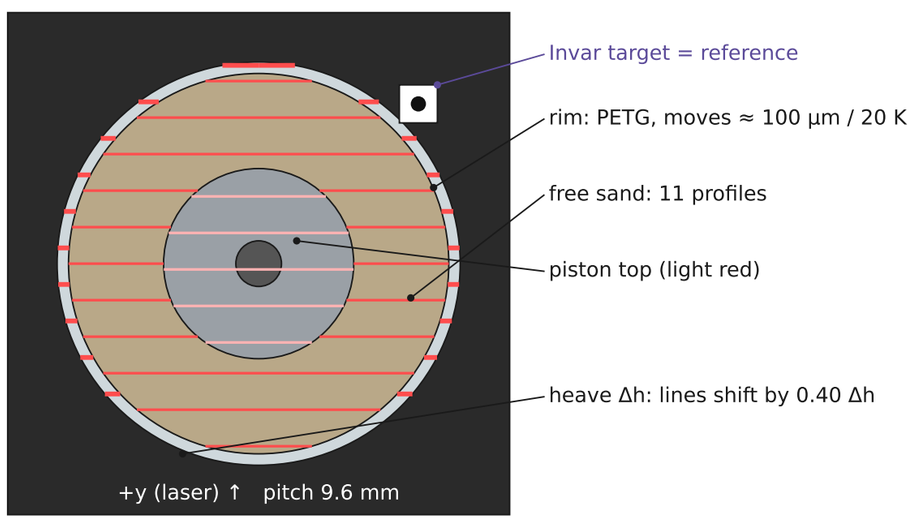
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**≈ 1 : 2.5**	drawing no.	**E7-V01**	sheet 1 / 6

(a) LASER TRIANGULATION – section y–z 1 : 1.25

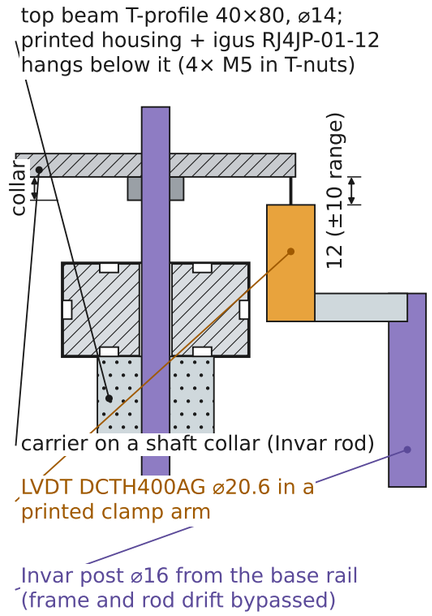


heave Δh moves the hit point by Δy = Δh / tan 68° = 0.40 Δh (open circle);
no collar shadow: the laser sits above the cavity edge

(b) CAMERA VIEW – 11 laser lines, whole surface



(c) LVDT ON INVAR POST 1 : 2.6



What is measured

method	signal	resolution	interval
(1) camera + 11 lines	map h(x, y) of the whole surface	≈ 10 µm	every 10 / 1 min
(2) LVDT	carrier = piston	≈ 1 µm	CR1000, 10 / 1 min
Invar target	camera / frame drift	≈ 10 µm	every photo

Evaluation per freeze–thaw cycle

quantity	how
surface map	Δh(x, y) on 11 profiles → contour map, whole ⌀100
heave, free surface	mean Δh over the annulus (minus rim)
heave under load	Δh_load = LVDT (check: (1) on the piston)
settlement on thawing	drop from the maximum to the end of thaw
residual per cycle	height at end of thaw − start of cycle
effect of the load	Δh_free − Δh_load
strain	Δh / 100 mm
frost depth	0 °C isotherm from Pt100 / fibre / CS645 (M)

Procedure

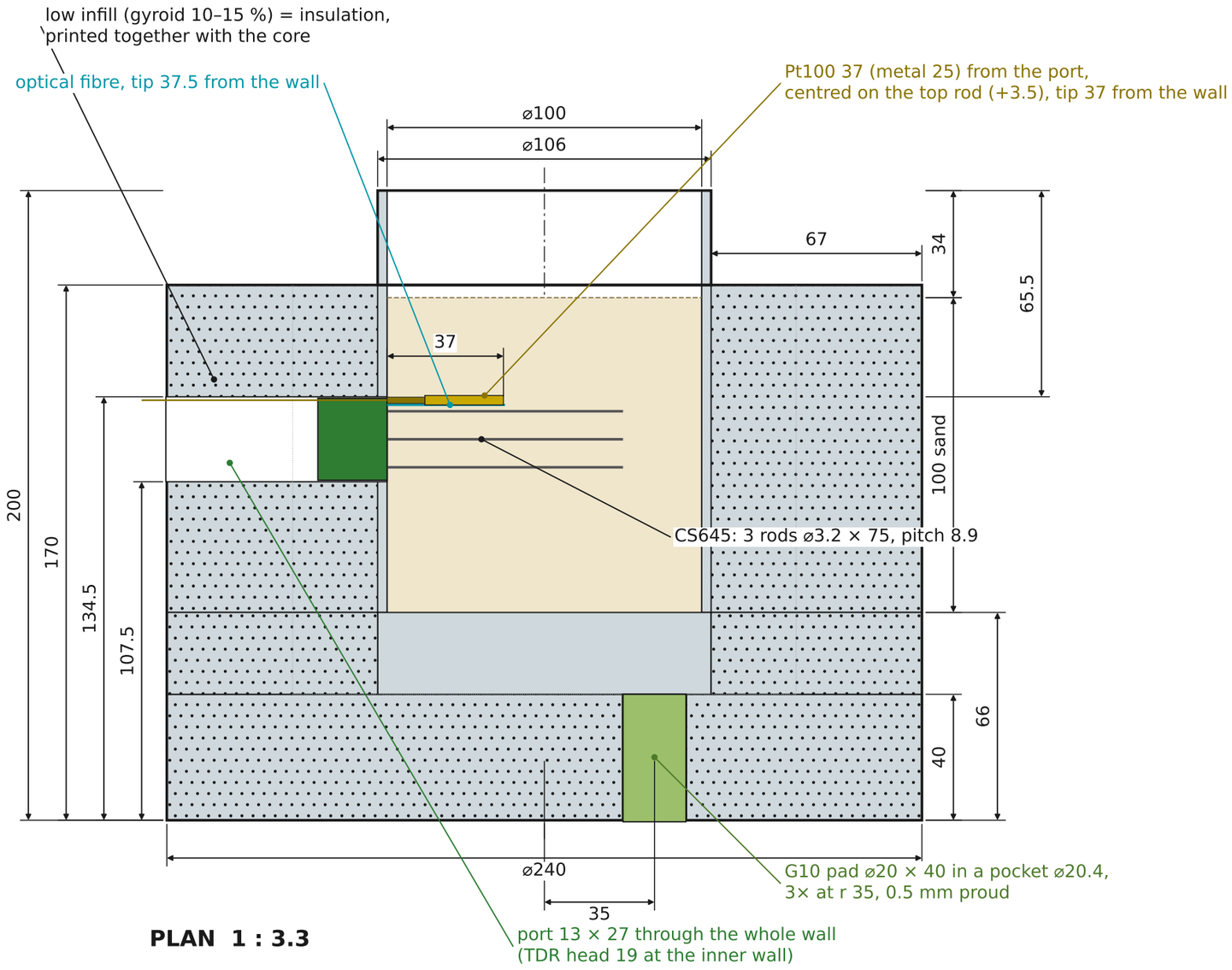
1 blank run, Al dummy instead of sand, full cycle: drift of (1) and (2) vs cell temperature (≈ 50 µm / 20 K, smooth) → fit and subtract
2 calibrate (1): shims 0.1 / 0.5 / 1.0 mm under a flat plate, camera px → mm for free sand and piston top
3 load, 24 h at +10 °C, then zero (1) and (2)
4 cycle 24 h −10 °C / 24 h +10 °C (fan on); CR1000 C5 pulse triggers the photo; laser on only for the photo
5 weigh the cell before / after each cycle; log air RH
6 expected heave for Sr ≈ 0.4 is small (tens of µm) – decide Sr first

heave while freezing, settlement while thawing, residual per cycle

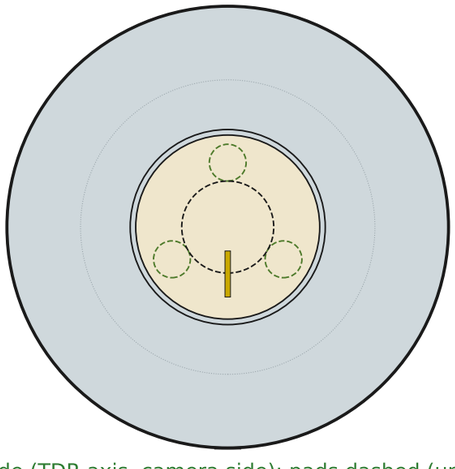
Deformation measurement – (1) camera + laser, (2) LVDT

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**various**	drawing no.	**E7-V02**	sheet 2 / 6

SECTION A–A 1 : 1.67 (along the TDR axis)



PLAN 1 : 3.3



port 13 wide (TDR axis, camera side); pads dashed (underneath)

Point M (existing sensors)

	sensor	position
TDR	Campbell CS645	3 rods 75, pitch 8.9, in the port, depth 36/45/54
OF	optical fibre	at the top rod, tip 37.5 from the wall
Pt100	37 mm (25 metal)	from the port, on top of the top rod, tip 37

Print (one piece)

item	value
size	ø240 × 200 mm – bed ≥ 250 × 250, Z ≥ 200
material	PETG or PETG-CF, 0.2 mm layers, ≈ 1.3–1.6 kg
walls	cavity wall ≈ 5 mm (≈ 8 perimeters), outer 3 perimeters
floor	z 40–66 under the cavity: 100 % (stiff, flat)
insulation	base z 0–40 and wall r 80–120: gyroid 10–15 %
pockets	3× ø20.4 × 40 from below, r 35
time	≈ 40–60 h (one job)

- Sand 100 mm (z 66–166), 34 mm free above (piston 20 + travel).
- The cell stands on the 3 G10 pads only (0.5 mm proud): the low-infill base insulates but carries no load → no creep in the LVDT chain.
- Heat model: λ infill ≈ 0.06 W/mK → freezes from the top, last unfrozen ≈ 13 mm above the floor; 95 % frozen in ≈ 13 h, thawed in ≈ 13 h.
- Inside: epoxy coat + PTFE spray + silicone grease.
- Cables: TDR head, Pt100 and fibre all through the port; the sand surface stays completely free (camera + laser).
- Full dimensions: cad/E7_V04_cell.dxf.

core per M. Vrána 23.06.2026 + 40 mm printed insulation; z = 0 at the underside

Printed cell rev. B – one piece ø240 × 200 with insulation, point M

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**1 : 1.67**	drawing no.	**E7-V03**	sheet 3 / 6

To buy

tag	item	EUR	CR1000
CAM	Raspberry Pi 5 4 GB + PSU, SD, camera cable (RPishop)	105–120 *	– (Ethernet, NTP)
+	Raspberry Pi HQ Camera C/CS, IMX477 (RPishop OFI060)	50–62 *	C5 trigger
+	Raspberry Pi 6 mm CS wide-angle lens (RPishop / Pi Hut)	20–30	–
+	camera box XPS 30, double glass; houses the RPi 5 (no heater)	20–40	–
+	SHT45 RH/T sensor on the RPi (I2C), freezer air	15–25	– (RPi)
LAS	Picotronic DOE253-635-1-6(10x22), 11 lines, 635 nm, 1 mW, 5 V	92 *	– (RPi GPIO)
LVDT	RDP DCTH400AG ±10 mm, spring (Elkome)	581 *	SE1, ±5000 mV
+	Mean Well HDR-15-24, 24 V DIN (floating supply) (TME)	11–14 *	C4 relay
+	Campbell 4WPB100 (for the existing Pt100) (Ekotechnika)	60–120	DIFF1
Tair	Campbell 109-L, REQUIRED: air = boundary condition	150–220	SE3, VX1
+	fan 80 mm 12 V, air mixing in the freezer	10–20	C4 relay
CAL	shims 0.1 / 0.5 / 1.0 mm (feeler gauge set) for camera px → mm	10–25	–

Mechanics (estimate)

item	EUR
laser: Al 10 base + top plate, carrier, piston discs; steel discs	100–190
profiles 2× 40×80 slot 8 × 326, 8 angle brackets, T-nuts, 3 feet M8	60–110
igus RJ4JP-01-12, shaft collar 12, ball 5/16", screws M3–M8	25–50
Invar 36: rod ⌀12 × 289 + post ⌀16 × 357 (Evek, cut to length)	80–160
cell printed in ONE piece ⌀240 × 200 (PETG(-CF) ≈ 1.5 kg) + 3 G10 pads ⌀20 × 5	45–90
printed parts: housing, arms, foot, laser holder, camera box (PETG-CF)	10–25

Total ≈ 1444–1974 EUR excl. VAT (≈ 36–49 tis. Kč). Already available: printed cell, CS645 + TDR100/200, Pt100, optical fibre, CR1000.

CR1000 channel plan

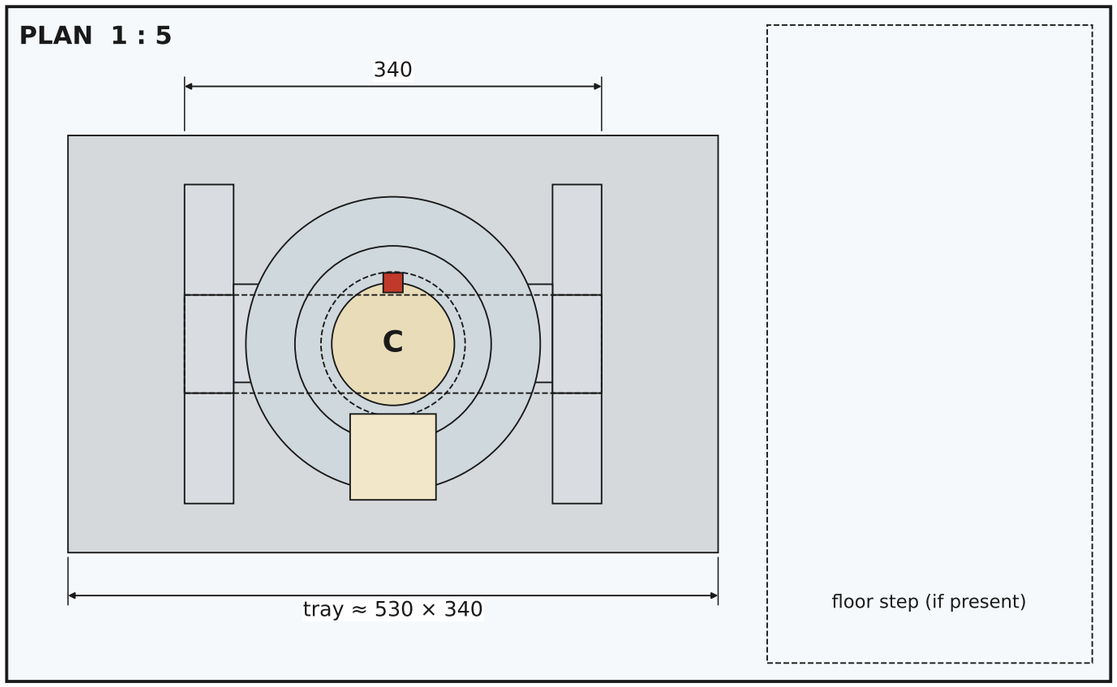
terminal	connected to	measurement
SE1	(2) LVDT	VoltSE ±5000 mV
DIFF1	Pt100 via 4WPB100	BrHalf4W, VX2 2500 mV
SE3	Tair 109 (required)	Therm109, VX1
C1–C3	TDR100/200 → CS645	SDM
C4	relay: LVDT supply	on for the scan
C5	pulse → RPi GPIO (opto)	triggers photo + laser
–	(1) camera + RH sensor	RPi 5 in the camera box
–	optical fibre	own reader

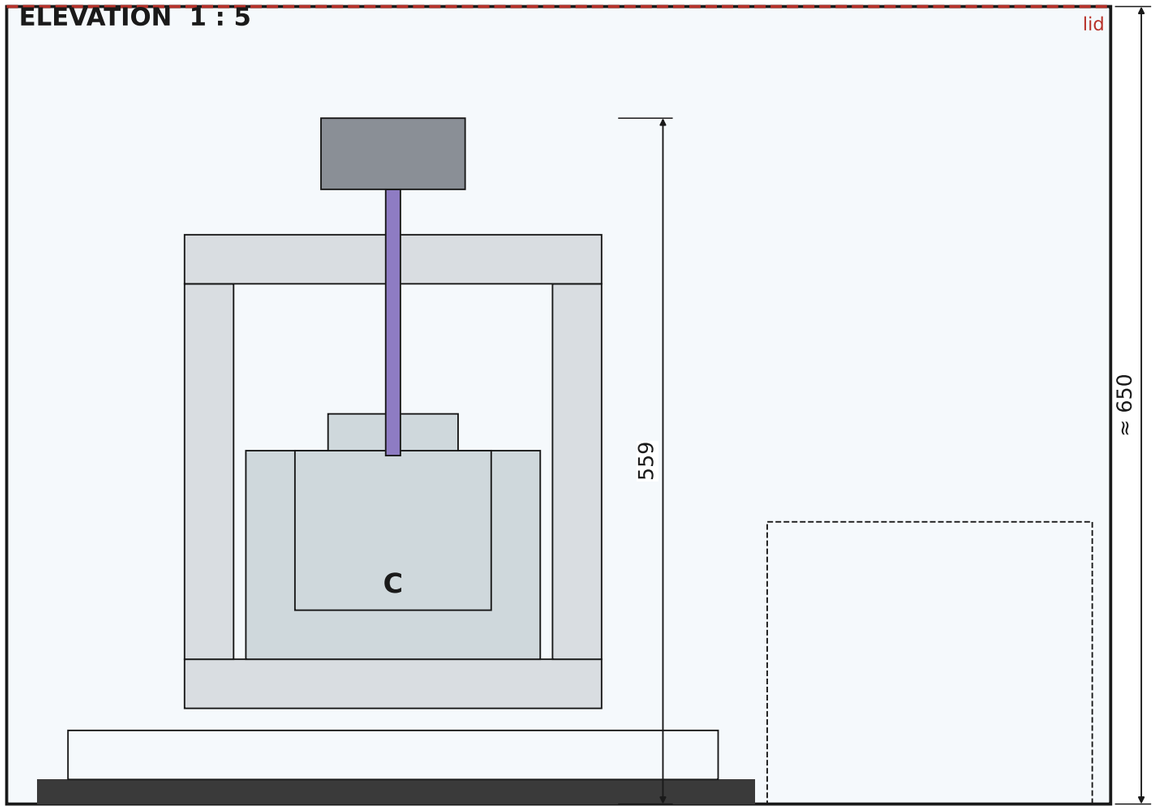
Notes
- CR1000 and TDR outside; RPi 5 inside the camera box (no CSI ribbon through the seal, its ≈ 5 W heat the box). Through the lid seal: Ethernet, 5 V, LVDT, sensor cables.
- Scan 10 min, 1 min around 0 °C; the photo is triggered by the CR1000 (C5) → same time stamps without NTP.
- Fan in the freezer evens out the on/off air swings.
- No pressure cells: the load is a known dead weight.

existing = already in the lab; prices ≈ EUR excl. VAT, * = dealer price found, rest estimated

Sensors, CR1000 wiring and costs

status	DRAFT	drawn Claude Code (draft)	checked: —
units	mm	project E7 – freeze–thaw loading tests	date 2026-11-09
scale	–	drawing no. **E7-V04**	sheet 4 / 6





Fit check

item	size	value	status
base frame	340 × 260	tray ≈ 530 × 340	OK
rig height	frame + discs	499 above tray	check
highest point	above freezer floor	559	≈ 91 spare
load travel	± 25 mm above the beam	max 584	lid closes?

Notes

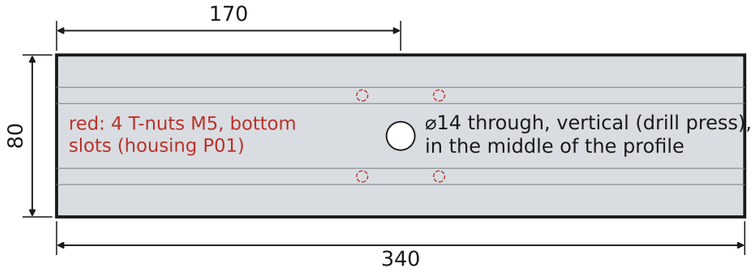
- The free end of the freezer stays empty (cable bundle, a second rig).
- Highest point is the disc stack; with 4 discs ≈ 50 mm under the lid – measure the freezer and the tray before ordering the profiles.
- Camera box and laser hang inside the frame – nothing outside the base.
- Rev. B3: frame of T-slot profiles; the bearing housing hangs under the top beam, beam gap 140 mm above the sand → the rig is lower than rev. B.
- 80 mm fan on the free end mixes the air; run the blank test first.

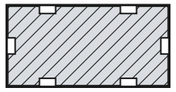
same freezer as E6 (≈ 900 × 550 × 650, to be measured); CR1000 outside the freezer

Rig C rev. B3 in the chest freezer

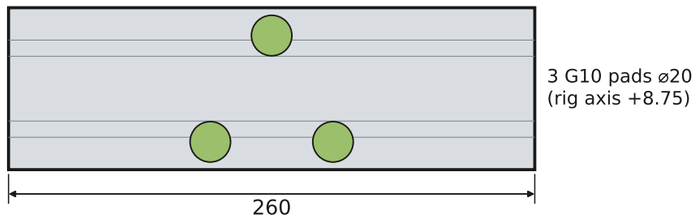
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**1 : 5**	drawing no.	**E7-V05**	sheet 5 / 6

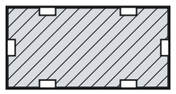
T04 TOP BEAM 40×80 × 340, 1×, lies flat



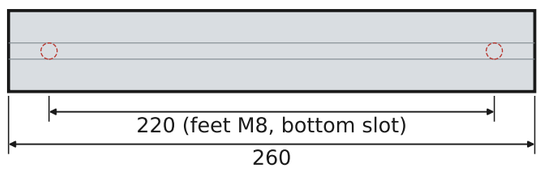


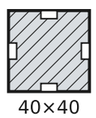
T03 BASE CROSS BAR 40×80 × 260, 1×, lies flat (axis y = 8.75)



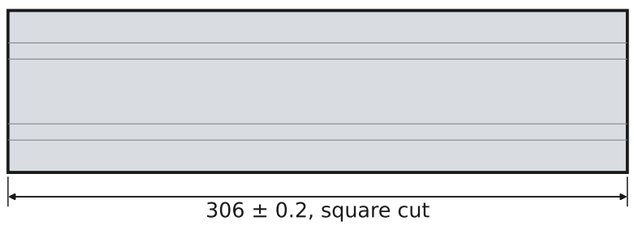


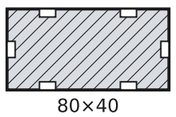
T02 BASE RAIL 40×40 × 260, 2×

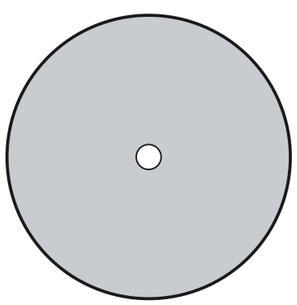


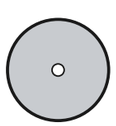


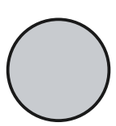
T01 COLUMN 40×80 × 306, 2×, both the same length

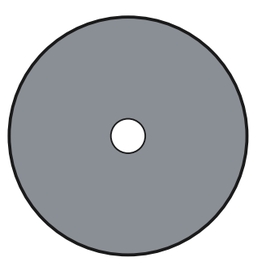


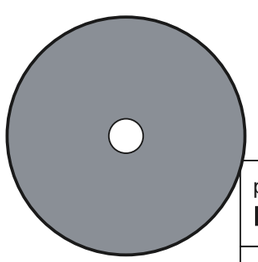












T-slot profiles slot 8 (cut to length, item/Bosch compatible)

#	section	length	pcs	part
T01	40×80	306	2	column
T02	40×40	260	2	base rail
T03	40×80	260	1	base cross bar
T04	40×80	340	1	top beam, ø14

Laser cut (one order, DXF 1 : 1)

#	size	t	material	pcs
L03	ø140	10	Al EN AW-5754	1
L04	ø50	10	Al EN AW-5754	1
L05	ø50	10	Al EN AW-5754	1
L06	ø117.5	12	steel S235JR	4
L07	ø117.5	6	steel S235JR	1

Cut bars

bar	length	pcs	note
Invar 36 ø12 h9	265 mm	1	guide rod, faces square, 0.5×45°
Invar 36 ø16	330 mm	1	post, faces square
G10 (sklotextit) ø20	40 mm	3	pads, faces parallel ±0.05

3D print (PETG-CF, 0.2 mm, 40 % gyroid unless noted)

#	part	pcs
V03	cell ø240 × 200, one piece (see V03)	1
P01	bearing housing	1
P02	lvdt arm	1
P03	target arm	1
P04	post foot	1
P05	laser holder	1
P06	camera box	1
P07	camera box lid	1
P08	camera hanger	1

Fixing: 12 angle brackets 40 slot 8 + 24 T-nuts M8 + M8×16 (4 base, 4 column–rail, 4 column–beam); 4 feet M8 in T-nuts; T-nuts M5: housing 4× M5×35, P04 / P05 / P08 2× M5 each; clamps M3/M4 × 20 + nuts. Use drop-in (swivel) T-nuts.

profiles only cut to length, one ø14 hole in T04; DXF in vyroba/laser, STL in vyroba/tisk

Parts to make – T-profiles, laser, bars, 3D print (rev. B3)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**1 : 3**	drawing no.	**E7-V06**	sheet 6 / 6